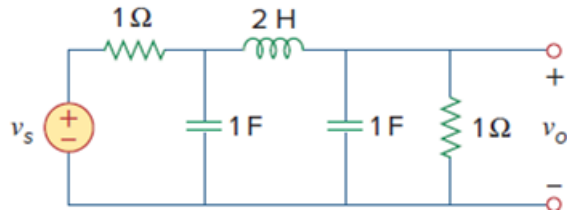


Problem

A third-order Butterworth filter normalized to $\omega_c = 1 \text{ rad/s}$ is shown in Fig. 14.49. Scale the circuit to a cutoff frequency of 10 kHz. Use 15-nF capacitors.

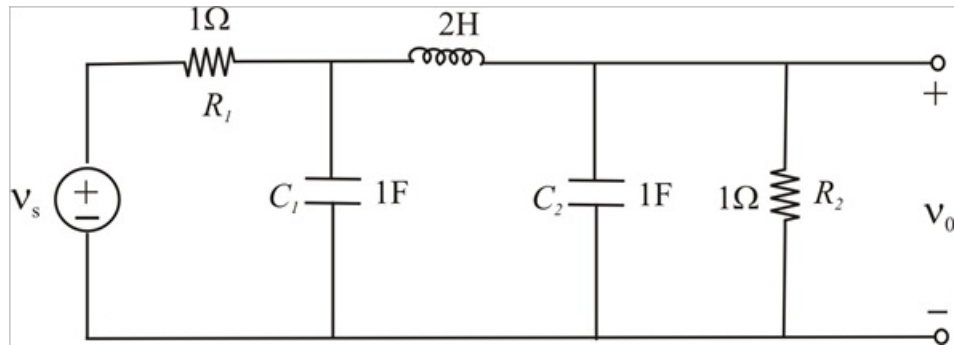
Figure 14.49:



Step-by-step solution

Step 1 of 7

Given circuit is,



Step 2 of 7

Given that,

$$\omega_c = 1 \text{ rad/sec}$$

$$f_c' = 10 \text{ kHz}$$

$$C' = 15 \text{ nF}$$

If the cut off frequency is to shift from

$$\omega_c = 1 \text{ rad/sec to } \omega_c' = 2\pi f_c' \text{ then}$$

$$\Rightarrow \omega_c' = 2\pi \times 10 \times 10^3$$

$$\therefore \omega_c' = 62.832 \text{ k rad/sec}$$

Step 3 of 7

The frequency scale factor is,

$$\Rightarrow K_f = \frac{\omega_c'}{\omega_c}$$

$$\Rightarrow K_f = \frac{62.832 \times 10^3}{1}$$

$$\therefore K_f = 62.832 \times 10^3$$

Step 4 of 7

Also, if each 1F capacitor is to be replaced by a 15 nF capacitor, then the magnitude scale factor must be

$$\Rightarrow C' = \frac{1}{K_m K_f} \times C$$

$$\Rightarrow K_m = \frac{1}{C' K_f} \times C$$

$$\Rightarrow K_m = \frac{1}{(15 \times 10^{-9}) \times (62.832 \times 10^3)} \times 1$$

$$\therefore K_m = 1.061 \times 10^3$$

Step 5 of 7

Therefore,

$$\Rightarrow R_1' = K_m R_1$$

$$\Rightarrow R_1' = 1.061 \times 10^3 \times 1$$

$$\therefore \boxed{R_1' = 1.061 \text{ k}\Omega}$$

$$\Rightarrow R_2' = K_m R_2$$

$$\Rightarrow R_2' = 1.061 \times 10^3 \times 1$$

$$\therefore \boxed{R_2' = 1.061 \text{ k}\Omega}$$

Step 6 of 7

$$C_1' = \frac{1}{K_m K_f} \times C_1$$

$$\Rightarrow C_1' = \frac{1}{(1.061 \times 10^3) \times (62.832 \times 10^3)} \times 1$$

$$\therefore \boxed{C_1' = 15 \text{ nF}}$$

$$C_2' = \frac{1}{K_m K_f} \times C_2$$

$$\Rightarrow C_2' = \frac{1}{(1.061 \times 10^3) \times (62.832 \times 10^3)} \times 1$$

$$\therefore \boxed{C_2' = 15 \text{ nF}}$$

Step 7 of 7

$$L_1' = \frac{K_m}{K_f} \times L_1$$

$$\Rightarrow L_1' = \frac{1.061 \times 10^3}{62.832 \times 10^3} \times 2$$

$$\therefore \boxed{L_1' = 33.77 \text{ mH}}$$